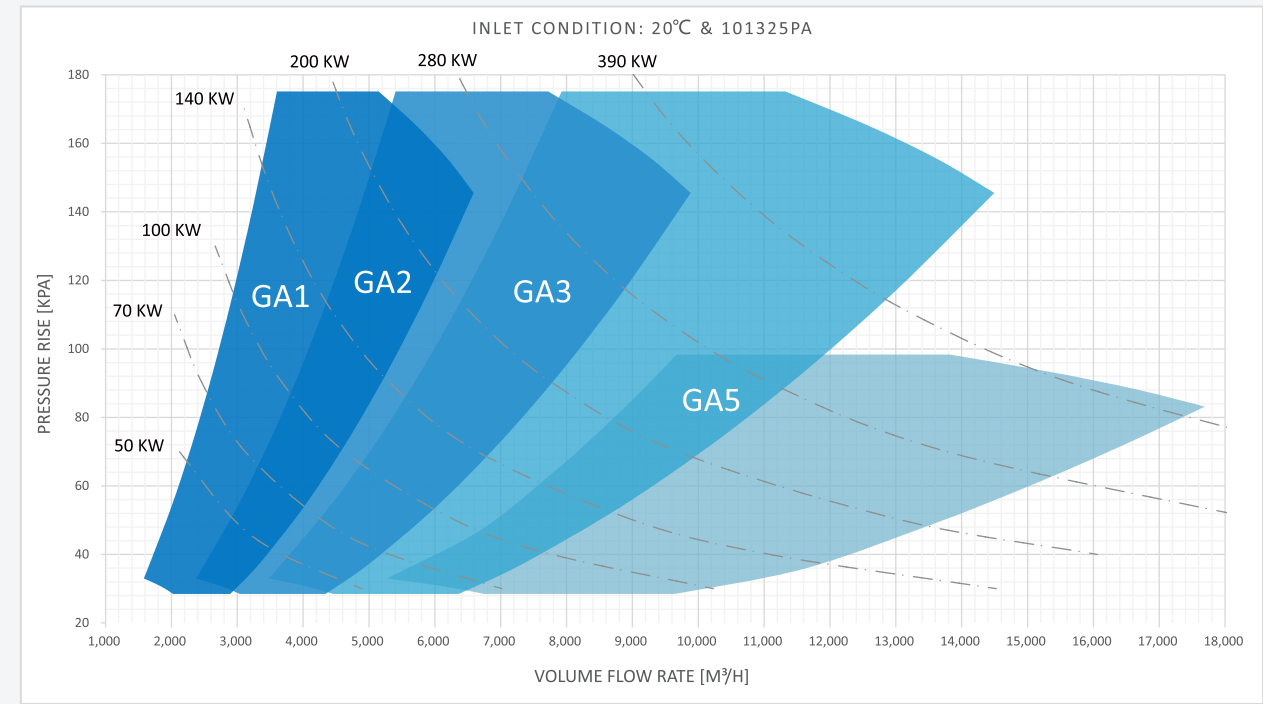


Product Specifications

Models

GL-TURBO customizes the most suitable blower for the actual operating conditions for customers worldwide. GA-Hybrid has the following models to meet our customers' various needs for stability, efficiency, flow regulation, and precision control.



Data Sheet

Model	Flow (m³/h)	Power (kW)	PD (kpa)
GA1	1600 – 6400	50 – 140	140
GA2	2400 – 9800	50 – 200	140
GA3	3600 – 14400	70 – 280	140
GA5	5400 – 17600	100 – 390	70

GL-TURBO

- ★ GL-TURBO Global Headquarters
- GL-TURBO Global Branches



www.glturbo.com
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GA-Hybrid Turbo Blower

Combines the best of geared and gearless turbos



Power
70 - 540 HP / 50 - 400 kW

Flow
940 - 10000 SCFM
1600 - 17000 m³/h

ΔP
4 - 25 PSI / 0.3 - 1.7 bar

GL-TURBO

GL-TURBO (GA-2401)_US_Headquarter 04 | 2024

GA-Hybrid Turbo Blower

Introduction

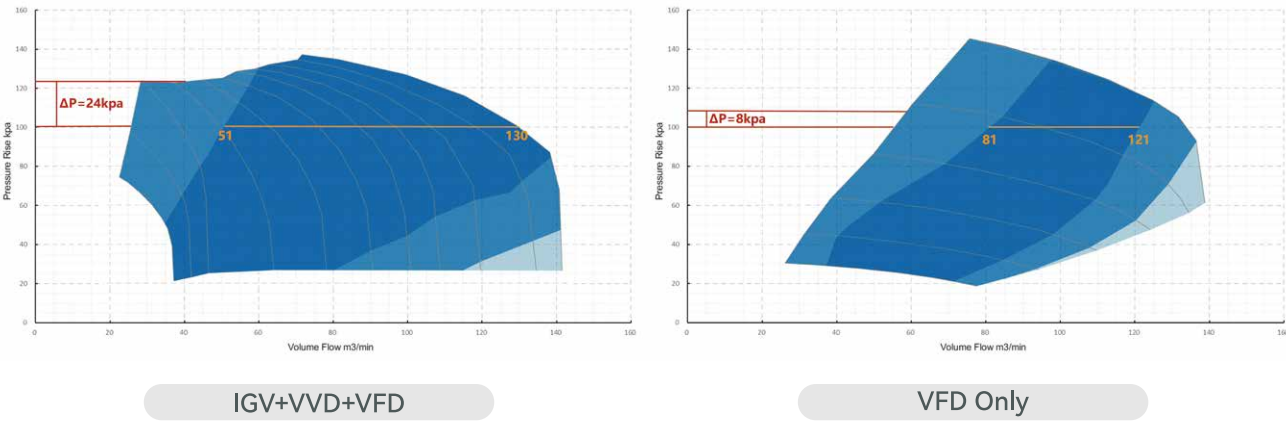
GA-Hybrid combines the best of geared and gearless turbo blowers, featuring great stability, wide flow range, high efficiency in entire flow range, small footprint, low noise, long service life, low maintenance cost and good performance price ratio, etc.

All these features are long desired by processes like wastewater aeration. The patented GA-Hybrid is the first blower in the world that meets the challenge successfully.

Key Features

1 Wider High Efficiency Area and Larger Surge Margin

Compared with VFD only turbo blowers, GA-Hybrid has much wider high efficiency range and much larger surge margin.



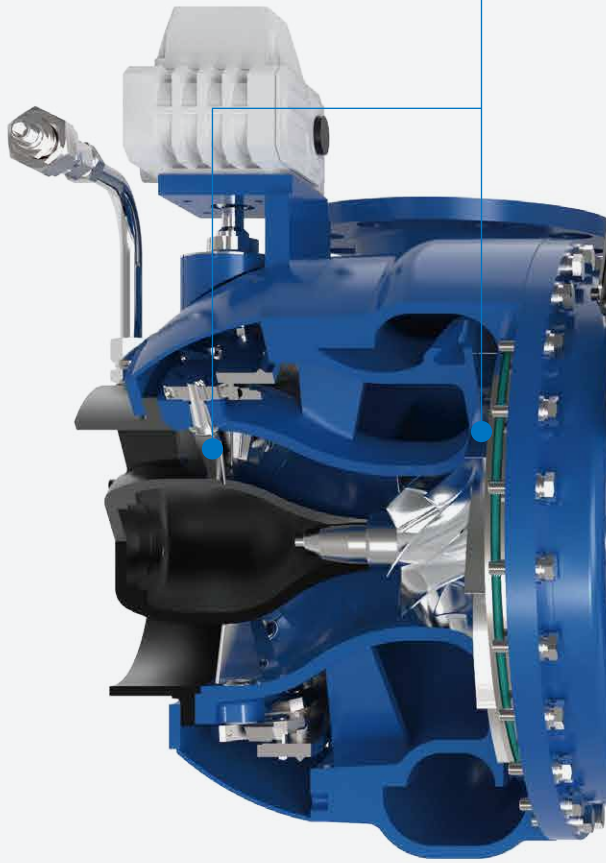
2 Stability & Surge Proof

Hydrodynamic oil films on bearings and gears have a load capacity of more than 3 times that of surge, safeguarding them and mating surfaces from surge damages.



1 Dual-Vane Control

Combined with VFD, IGV and VVD provide a wide flow range (more than 35%-105%) and high efficiency throughout.



2 Hydrodynamic Tilt-Pad Journal Bearing

Installed on high-speed pinion. No direct contact with pinion shaft, resulting in a long service life of over 100,000 hours.

The load capacity is more than 3 times that of surge, safeguarding bearing and rotor from damage.

3 Thrust Collar

Installed on both sides of pinion gear. Transfers thrust load to low-speed shaft via bull gear. Thus greatly reduces overall bearing loss.

4 Gear Driven

Proven stability and long service life.

No direct contact between bull gear and pinion gear during operation, resulting in very little power loss.

5 Motor Primary Bearing

Installed on motor shaft, this combined journal and thrust hydrodynamic bearing carries both radial and axial load for the motor.

No direct contact between bearing and mating surfaces, resulting in a long service life of over 100,000 hours.

The load capacity is more than 3 times that of surge, safeguarding bearing and mating surfaces from damage.

6 Permanent Magnet (PM) Motor

Proven technology that operates at 6,000 rpm vs 20,000.

Smaller size and higher efficiency at partial load than induction motor.

Mounted on the gearbox directly through flanges, with bull gear installed on motor rotor, resulting in a more compact and efficient system.

7 Variable Frequency Drive (VFD)

Proven technology that drives a PM motor at 6,000 rpm vs 20,000 rpm.

Combined with IGV and VVD, provides a 3rd dimension for blower control that results in a wide flow range and high efficiency throughout.

Adjusts motor speed to accommodate standardized gear ratios. Thus greatly reduces manufacturing cost and shortens lead time.

Adjusts motor speed to change discharge pressure to better match operating requirement, especially valuable when pressure of target system changes over time.

